MODULE 1 · TOPIC 01

# Introduction to Flora Surveys

*The first step of a flora survey: what it is, the pre-field preparation it depends on, and how the BAM, the law and the consent authority frame it.*

**DOCUMENT CONTROL & REVIEW**

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| 1.0 | — | Initial topic note. |
| 2.0 | Aug 2026 | Refocused on the first survey step & pre-field process; added legislative/consent-authority context, five-perspective analysis and deliverable standard. |

**HOW TO READ THIS DOCUMENT** Process / how-to · Legislative / assessment · Definitions / concepts · Pitfalls / caution · Perspectives

### LEARNING OBJECTIVES

- Explain what a flora survey is and the questions it answers
- Describe where the survey sits and what pre-field steps must come first
- Understand the BAM, legislative and consent-authority context that frames it
- Know what the pre-field stage must deliver before anyone goes to site

## GOVERNING DOCUMENTS & LEGISLATIVE CONTEXT

A flora survey is not free-form data collection — it is the field step of a method set by law. These instruments govern it:

- **Biodiversity Conservation Act 2016 (BC Act)** — the head of power; protects biodiversity, lists threatened species and TECs, and establishes the Biodiversity Offsets Scheme.
- **Biodiversity Conservation Regulation 2017** — gives the BAM legal force and sets the offsets-scheme entry thresholds.
- **Biodiversity Assessment Method (BAM 2020)** — the governing method your survey must satisfy: what to record, how much, and to what standard.
- **EPBC Act 1999 (Cth)** — applies in parallel where Matters of National Environmental Significance may be affected.
- **Threatened-species survey guidelines (NSW & Cth)** — set the "adequate survey effort" your method is judged against.

Cite the instrument and version in your field plan and notes — e.g. "survey designed per BAM 2020; effort per NSW survey guidelines".

## HOW THE LEGISLATION LINKS TO YOUR SURVEY

The law does not describe plants — it describes a chain of obligations, and the flora survey is the evidence step that feeds it. Reading the chain tells you why the survey must be done a particular way.

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| **The Act sets the duty** | The **BC Act** makes it an offence to unlawfully harm biodiversity and requires impacts to be assessed. It is *why* a survey happens at all. |
| **The Regulation triggers assessment** | The **BC Regulation** sets the offsets-scheme thresholds (Biodiversity Values Map, area cleared) that decide whether a formal BAM assessment is needed — and gives the BAM legal force. |
| **The Method sets the survey** | The **BAM** prescribes how the survey is done — stratify into zones, run plots, record the required attributes and target species. This topic is the front door to that method. |
| **The guidelines set the effort** | The **survey guidelines** define “adequate effort” — coverage, replication and timing. Meet them and the survey is defensible; miss them and it can be rejected. |
| **The consent authority tests it** | DPE / council relies on your survey to make a lawful decision. If effort or method fall short, they issue a Request for Information (RFI) — delay and cost for the client. |

**The link in one line:** BC Act (duty to assess) → BC Regulation (does the BOS apply?) → BAM (how to survey) → survey guidelines (how much effort) → your field data → the report the consent authority relies on. **The survey is the evidence the whole chain stands on.**

## DEFINITIONS — KEY TERMS

**Flora survey** — a systematic field record of the plant species and vegetation present at a site.

**Desktop review** — pre-field compilation of maps, imagery and database records that predicts what may occur and frames the survey.

**Scope of work (SoW)** — the agreed statement of what the survey will and will not cover, its method, effort and deliverables.

**Field plan** — the practical plan for the survey: target list, method, timing, access, safety and logistics.

**Adequate survey effort** — enough coverage, replication and correct timing to reliably detect what is present.

**Traceability** — every reported figure being traceable back to a field record.

## BACKGROUND & WHY IT MATTERS

A flora survey is a structured record of the plants and vegetation at a site. We do them because planning and environmental decisions must know what lives on the land — the native vegetation present, any threatened plants or communities, and the condition of the vegetation.

A survey is only as good as its preparation. Before anyone goes to site, the **desktop review, scope of work and field plan** set what you are looking for, how, and to what standard. Skip these and you risk the wrong timing, inadequate effort, or a survey that the consent authority will not accept. This topic is deliberately about that first step — getting the survey framed correctly — not about the later report.

## THE TOPIC FROM FIVE PERSPECTIVES

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| **Legal** | The survey is the evidence that discharges the duty to assess impacts under the BC Act (and EPBC Act). An inadequate survey can make a later approval legally vulnerable. |
| **Assessment** | The BAM dictates what the survey must capture. Your pre-field planning ensures the method and effort will produce valid BAM inputs — not data that has to be re-collected. |
| **Consent authority** | Reviewers judge whether survey effort and timing were adequate. A clear scope and field plan, followed and documented, are what let them accept the work without an RFI. |
| **Client / stakeholder** | Good preparation controls cost and timing risk. Getting the season and effort right first time avoids repeat mobilisations and approval delays — the client's two biggest exposures. |
| **Our consultancy (unbiased)** | We scope surveys to what is required to find the truth about a site — no less to please a budget, no more to pad a fee. Our credibility rests on independent, adequate survey design. |

## THE PRE-FIELD PROCESS — STEP BY STEP

Everything in this topic happens *before* boots hit the ground. Each step has a purpose and a product.

**Step 1 — Review the desktop search** (why: predicts what may occur and frames effort)

1. Read the desktop research outputs (BioNet, EPBC PMST, mapping) to build the list of PCTs and threatened species that could occur.
2. Note each target species' detectability window so the survey can be timed correctly. Product: a target list with seasons.

**Step 2 — Confirm the scope of work** (why: sets what the survey must deliver)

3. Read the scope/proposal to confirm the objectives, area, method and deliverables, and which law/method applies.
4. Raise any gap or ambiguity with the Project Manager before mobilising. Product: a confirmed SoW.

**Step 3 — Build the field plan** (why: makes the survey adequate, safe and efficient)

5. Choose the method(s) and effort to match the objectives and the survey guidelines; set the survey timing to the target windows.
6. Plan access, landholder permissions, licences, safety/SWMS and equipment. Product: a field plan ready to execute.

**Step 4 — Confirm & brief** (why: everyone mobilises to the same plan)

7. Confirm the plan with the Project Manager/SME and brief the field team on targets, method, roles and safety before departure.

**Hands over to:** Topic 02 (Objectives), Topic 04 (Methodologies) and the field-execution topics. This topic ends when the survey is correctly framed and the team is ready to mobilise.

## HOW TO RESEARCH & FIND THE INFORMATION

- Project folder — brief, scope of work, proposal and client information.
- Desktop research outputs — BioNet buffer search and EPBC PMST results, vegetation mapping.
- The BAM and relevant survey guidelines for method and adequate effort.
- Previous reports for the site or nearby land for precedent and known values.

## DELIVERING THE OUTCOME — WHAT SUCCESS LOOKS LIKE

The deliverable of this first step is a **survey that is correctly framed before mobilising** — a confirmed scope, a target list with timing, and a field plan that will meet BAM and guideline requirements. Success means:

- The survey is **designed to adequate effort and correct timing**, so it can withstand consent-authority review without an RFI.
- The method will produce **valid BAM inputs** — no data has to be re-collected.
- The plan is **safe, legal (access, licences) and efficient**, avoiding repeat mobilisations.
- We have scoped the survey **independently and to what the site actually requires** — documenting the facts, not a preferred outcome.

**COMMON PITFALLS**

- Mobilising before reading the desktop search, so targets or timing are missed.
- An unclear scope — surveying the wrong area or to the wrong standard.
- Booking the survey in the wrong season for the target species.
- Overlooking access, licences or safety planning until on site.

**CHECK YOUR UNDERSTANDING**

- What three pre-field products must exist before you mobilise?
- How does the BAM link back to the BC Act and forward to the consent authority?
- Why does the desktop search drive the timing of a survey?
- What does “adequate survey effort” mean, and who judges it?

**Related reading:** the **Desktop Research Guide** for the pre-field database work this topic builds on, and Topics 02 & 04 for objectives and methods.